

SERVICE NOTES

May 13, 2003

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FILE TITLE

Signa[®] System

SN 67086

RE-FLASHING SIGNA EXCITE SCP BOARD NVRAM IN THE FIELD

EFFECTIVITY:

All 10.0 EXCITE MR System SCP Boards (part number 2294300-2) used inside the MGD Chassis located in the System Cabinet.

PROBLEM:

Many failed SCP Boards (located in slot 11 on the front of the MGD Chassis) that have been returned from the field because the MGD would not download or boot have been found to have Non-Volatile RAM (NVRAM) contents that are corrupted. An SCP Board with corrupted NVRAM contents will prevent a successful TPS Reset from occurring.

SOLUTION:

The SCP Board NVRAM can be re-flashed in the field. Try performing the steps below before attempting to replace the SCP Board.

1. Open a C-shell window.
2. Login to the SCP Board by typing the commands as shown below.

```
t20 6# cu scp <Enter>
Connected
<Enter>
EPPC-Bug>
```

3. If the **EPPC-Bug>** prompt is seen after logging in then the SCP Board NVRAM is corrupted. Proceed to step 4. Otherwise, the SCP Board NVRAM is **not** corrupted and the SCP Board is likely not the source of the problem. Performing the re-flash process described in step 4 would not solve the problem.



Work slowly and double-check your entries while performing the process described in step 4. Do not power cycle the MGD until you are sure that all the values are correct. If any are wrong, start from the beginning and repeat the process again. If you cycle the MGD power with a wrong NVRAM entry the SCP may hang. The only way to recover will be to clear the NVRAM. See the end of this Service Note for directions on how to clear the NVRAM.

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4. Make the entries in bold print as shown below.

```
EPPC-Bug>ENV;D <Enter>
Update with Auto-Configuration Defaults

Update Non-Volatile Memory (Y/N)? Y <Enter>

Reset Local System (CPU) (Y/N)? N <Enter>
WARNING: Updates will not be in effect until a RESET is performed.
EPPC-Bug>ENV <Enter>
Probe System for Supported I/O Controllers [Y/N] = Y? <Enter>
Local SCSI Bus Reset on Debugger Startup [Y/N] = N? <Enter>
PCI Interrupts Route Control Registers (PIRQ0/1/2/3) = 0A0B0E0F? <Enter>
Firmware Command Buffer Offset = 000002C8? <Enter>
Firmware Command Buffer Size = 00000200? <Enter>
Firmware Command Buffer Delay = 5000? <Enter>
Program Intermediate Load Address = 00200000? <Enter>
Binary Program Load Address = 00080000? <Enter>
Binary Program Execution Offset = 00000100? <Enter>
Primary Network Controller LUN = 20? <Enter>
Primary Network Device LUN = 00? <Enter>
Firmware Command Buffer :
['NULL' terminates entry]? GO FC000100 <Enter>
SD <Enter>
SE;AEM;QST;PCIB;FECETH <Enter>
GO FC00017C <Enter>
NULL <Enter>

Update Non-Volatile Memory (Y/N)? Y <Enter>

Reset Local System (CPU) (Y/N)? N <Enter>
WARNING: Updates will not be in effect until a RESET is performed.
EPPC-Bug>
```

5. The safest way to get the SCP Board out of debug mode is to cycle power. Open the rear door of the EXCITE System Cabinet and locate the power switch (near the incoming AC power connection) on the left, rear of the MGD Chassis.
6. Place the MGD Chassis power switch in the down (OFF) position and wait for about 5 seconds before placing it back into the up (ON) position.

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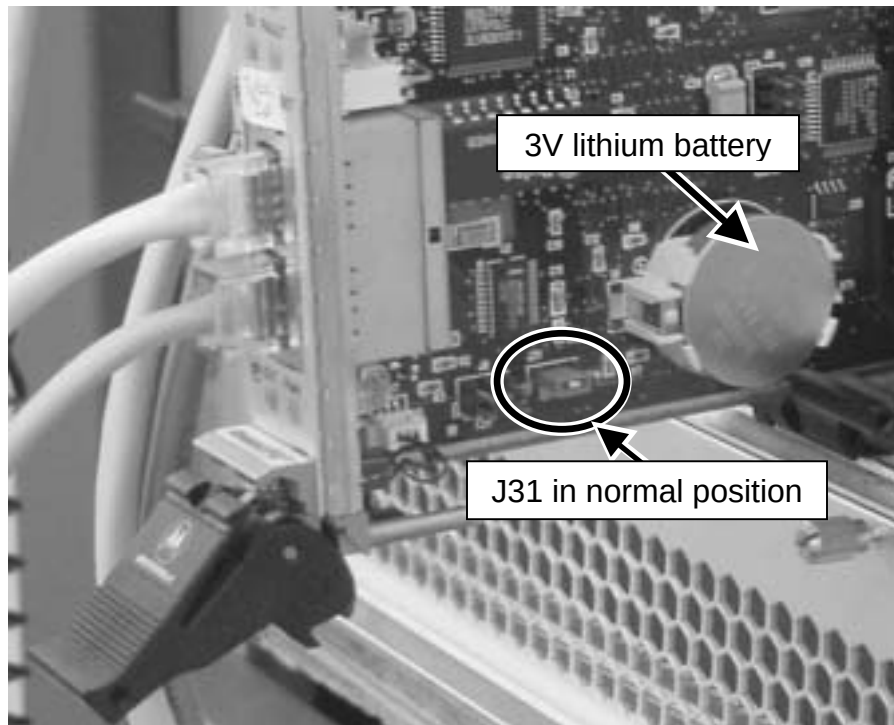
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7. Perform a **TPS Reset** from under *Utilities* on the MR Service Desktop.

If the C-shell window is open, and there is still a connection to the SCP Board, you should see information scrolling down the screen during the TPS Reset.

8. If the TPS Reset was successful then close the C-shell window to terminate the communication link to the SCP Board.

Once the TPS Reset is successful, perform a scan to confirm that the system is operating correctly. If it is not successful and the SCP Board is back in debug mode, repeat the process outlined in step 4 again. If the SCP Board still fails then power the MGD Chassis off and pull the SCP Board partially out of the slot as shown in Illustration 1. Locate the jumper J31 and the button battery on the bottom, front of the board.



JUMPER J31 AND BUTTON BATTERY LOCATIONS ON SCP BOARD
ILLUSTRATION 1

- Confirm that jumper J31 is across the 2 pins closest to the front faceplate of the board. This is the normal position for J31.
- Confirm that the button battery is firmly seated in its holder. Use a digital multimeter to measure the battery voltage (should measure about 3.1 VDC and not less than 2.5 VDC) and confirm that it is not too low. The exposed, flat side of the battery is the (+) terminal while the side of the battery is the (–) terminal.

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- Make certain that there are not any bent pins on the MGD Chassis mid-plane in the slot that contains the SCP Board or, for that matter, in any other slot.
- If you need to clear the NVRAM move the jumper at J31 momentarily from the left position to the right position and then back again to the left position. The jumper must remain in the left, normal position during normal operation.

When finished inspecting the SCP Board reseal it back into the MGD Chassis slot. If the NVRAM was cleared then repeat the process in step 4 again to re-flash the NVRAM.

If the jumper is the in the correct position, the battery is seated properly, and the battery voltage is OK but the SCP Board still won't download, then replace it.

Engineering is working with the vendor of this part to understand the root cause of the NVRAM corruption problem and to prevent it from happening.

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MR SERVICE ENGINEERING